

Isaac S. Garee

Cincinnati, OH gareeis@mail.uc.edu
isgaree.github.io | github.com/isgaree | linkedin.com/in/isgaree

Education

University of Cincinnati

B.S. Industrial and Systems Engineering
Minors in Mathematics and Statistics

Cincinnati, OH
Expected May 2027
GPA: 3.51/4.00

Graduate Coursework: Stochastic Processes and Applied Probability; Nonlinear Optimization

Selected Coursework: Linear Algebra; Deterministic Systems Models; Probabilistic Systems Models; Quality and Reliability Engineering; Probability and Statistics II; Data Science

Research Focus

Stochastic optimization and applied probability, with interests in bilevel and derivative-free optimization, optimization under uncertainty, and stochastic systems.

Research Experience

University of Cincinnati

Undergraduate Researcher
Stochastic Mixed-Integer Bilevel Optimization
Advisor: Prof. Tommaso Giovannelli

Cincinnati, OH
Apr. 2025 – Present

- Developed stochastic bilevel optimization methods for problems with mixed-integer upper-level decisions and incomplete derivative information.
- Designed and implemented a stochastic bilevel optimization framework combining derivative-free search, finite-difference information, lower-level learning, and stochastic acceptance and line-search mechanisms.
- Conducted computational studies on neural architecture search, comparing convergence, stability, architecture quality, and validation performance against established bilevel optimization baselines.
- Analyzed theoretical properties and algorithmic behavior of mixed-integer stochastic bilevel methods; resulting manuscript is in preparation.

University of Southern California

Data Science and Operations Summer Scholar
Stochastic Assembly Systems
Advisors: Prof. Andrew Daw and Prof. Vishal Gupta

Los Angeles, CA
May 2026 – Aug. 2026

- Studied stochastic multi-component assembly systems with finite inventory buffers using simulation and mathematical analysis.
- Built scalable Python simulations to characterize steady-state behavior across component counts and buffer capacities, enabling experiments in large-system regimes.
- Investigated asymptotic scaling laws for the steady-state number of missing component types, developing computational evidence for general buffer capacities beyond previously studied special cases.

Presentations

A Stochastic Algorithm for Bilevel Optimization with a Mixed-Integer Upper-Level Problem and Applications to Neural Architecture Search

INFORMS Annual Meeting, San Francisco, CA

Nov. 2026 (forthcoming)

A Stochastic Algorithm for Bilevel Optimization with a Mixed-Integer Upper-Level Problem and Applications to Neural Architecture Search

Modeling and Optimization: Theory and Applications (MOPTA), Lehigh University, Bethlehem, PA

Aug. 2026

The Effect of Storage Capacity in Large-Scale Assembly Systems

USC Data Science and Operations Summer Scholars, Los Angeles, CA

Aug. 2026

Manuscripts

Garee, I., Giovannelli, T.

A Stochastic Algorithm for Bilevel Optimization with a Mixed-Integer Upper-Level Problem and Applications to Neural Architecture Search.

Manuscript in preparation.

Honors & Fellowships

Selected Participant, GradOpt, Lehigh University

Aug. 2026

- Selected for the fully funded Future Optimization Graduate Student Workshop for prospective optimization Ph.D. students.

Undergraduate Research Co-op Fellowship (URCF), University of Cincinnati

May 2025

- Supported faculty-mentored undergraduate research in stochastic bilevel optimization.

Academic Service

Session Co-Chair, MOPTA, Lehigh University

Aug. 2026

- Co-chaired “Stochastic Optimization for Hierarchical, Distributed, and Robust Decision-Making and Learning I–II.”

Professional Experience

Kaivac

Hamilton, OH

Industrial & Systems Engineering Co-op

Jan. 2024 – Dec. 2024

- Reduced inbound inspection process time by 20% through value-stream mapping and implementation of process improvements.
- Developed Power BI dashboards to track manufacturing and quality metrics and improve visibility into operational performance.
- Applied statistical process control and control charts to manufacturing data and expanded data-collection and tracking processes.

Technical Skills

Programming: Python, R, PyTorch, NumPy, SciPy, Pandas

Optimization: Gurobi

Tools: Git/GitHub, L^AT_EX, Jupyter, Excel, Power BI

Leadership

Secretary, Industrial & Systems Engineering Student Association

Jan. 2024 – Jan. 2025

- Coordinated activities and communications for a 50+ member student organization.